

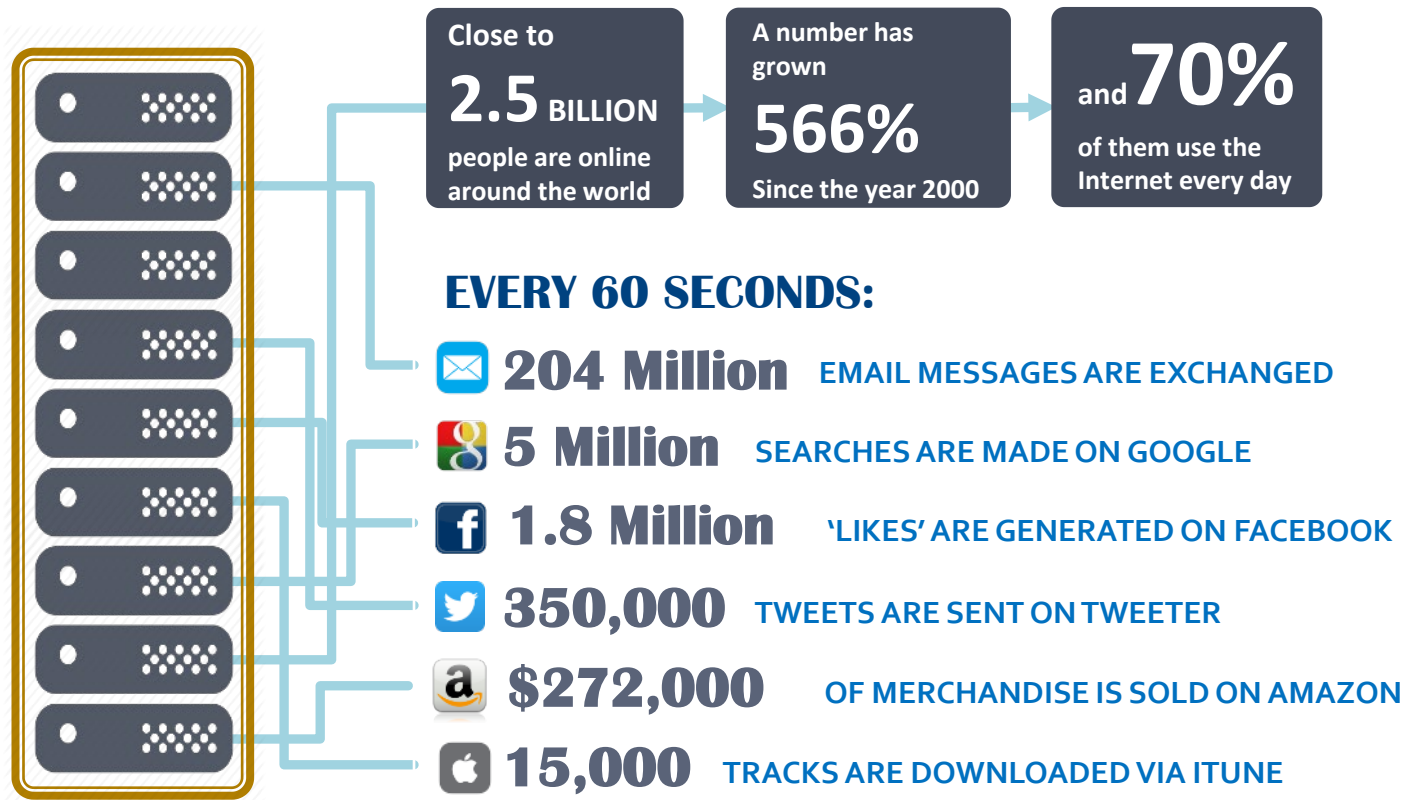
Big data processing & mining

- Introduction -

Dongchul Park

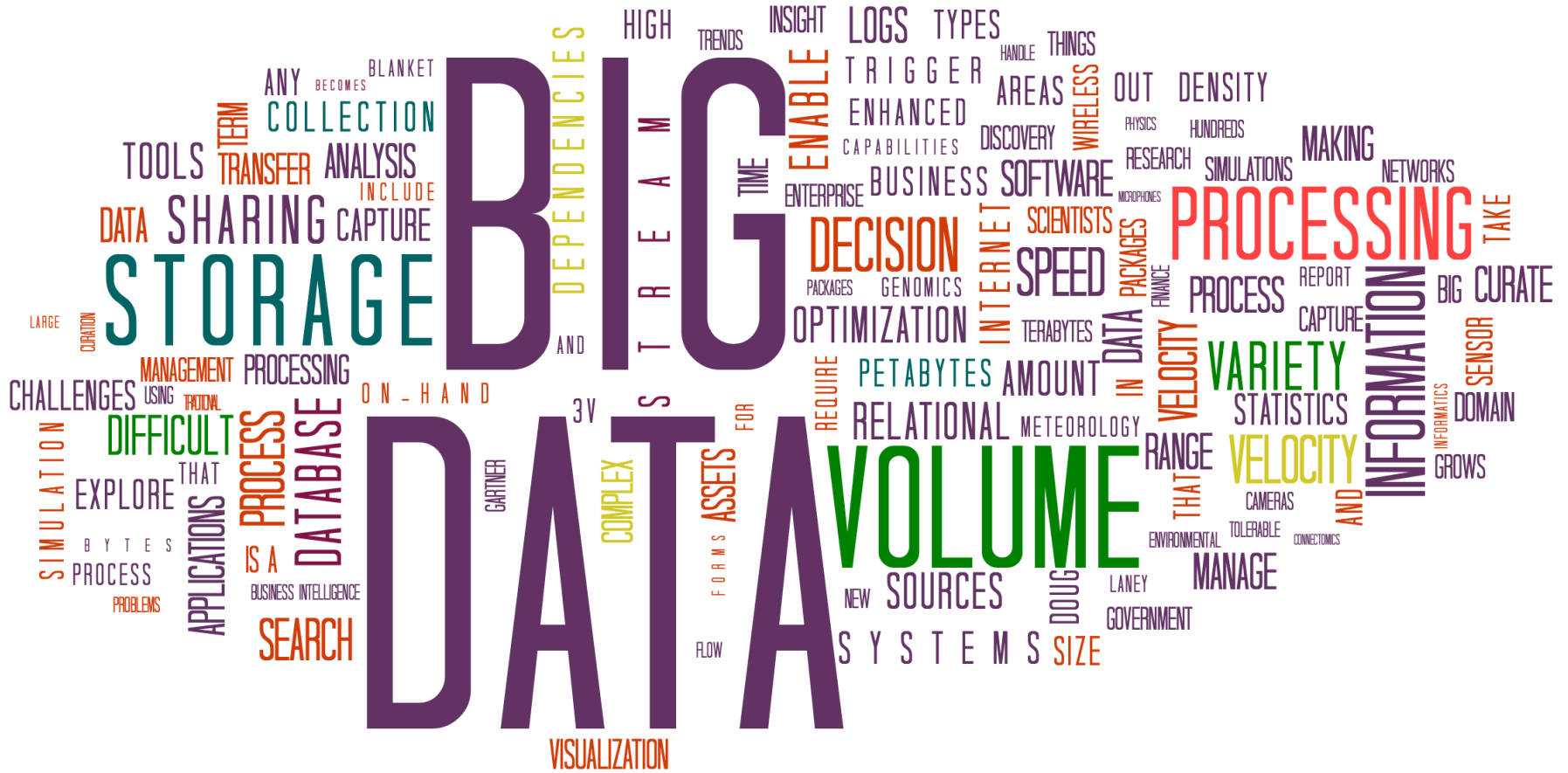
Chung-Ang University
Big Data Platforms and Security

Data Deluge

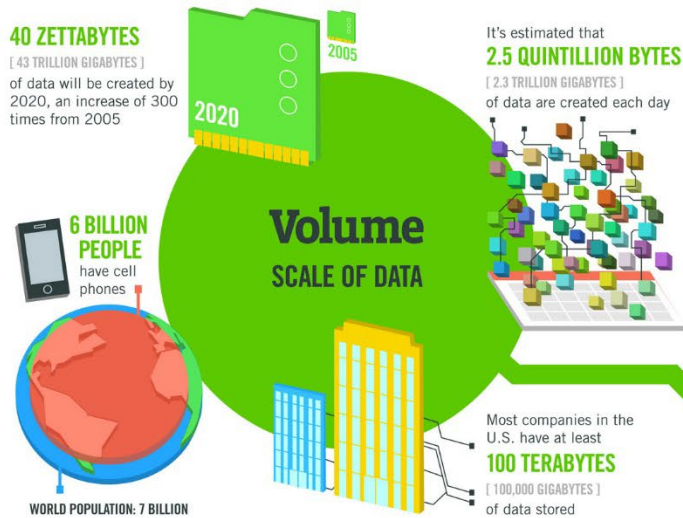


Source: NRDC, Data Center Efficiency Assessment

Big Data Era



4 Vs of Big Data



The FOUR V's of Big Data

From traffic patterns and music downloads to web history and medical records, data is recorded, stored, and analyzed to enable the technology and services that the world relies on every day. But what exactly is big data, and how can these massive amounts of data be used?

As a leader in the sector, IBM data scientists break big data into four dimensions: **Volume, Velocity, Variety and Veracity**

Depending on the industry and organization, big data encompasses information from multiple internal and external sources such as transactions, social media, enterprise content, sensors and mobile devices. Companies can leverage data to adapt their products and services to better meet customer needs, optimize operations and infrastructure, and find new sources of revenue.

By 2015
4.4 MILLION IT JOBS
will be created globally to support big data,
with 1.9 million in the United States

As of 2011, the global size of data in healthcare was estimated to be

150 EXABYTES
[161 BILLION GIGABYTES]



30 BILLION PIECES OF CONTENT
are shared on Facebook every month



Variety
DIFFERENT FORMS OF DATA

By 2014, it's anticipated there will be

420 MILLION WEARABLE, WIRELESS HEALTH MONITORS

4 BILLION+ HOURS OF VIDEO
are watched on YouTube each month



400 MILLION TWEETS
are sent per day by about 200 million monthly active users



The New York Stock Exchange captures
1 TB OF TRADE INFORMATION
during each trading session



Velocity
ANALYSIS OF STREAMING DATA

Modern cars have close to **100 SENSORS**
that monitor items such as fuel level and tire pressure



By 2016, it is projected there will be

18.9 BILLION NETWORK CONNECTIONS

— almost 2.5 connections per person on earth



1 IN 3 BUSINESS LEADERS

don't trust the information they use to make decisions



27% OF RESPONDENTS

in one survey were unsure of how much of their data was inaccurate

Veracity
UNCERTAINTY OF DATA

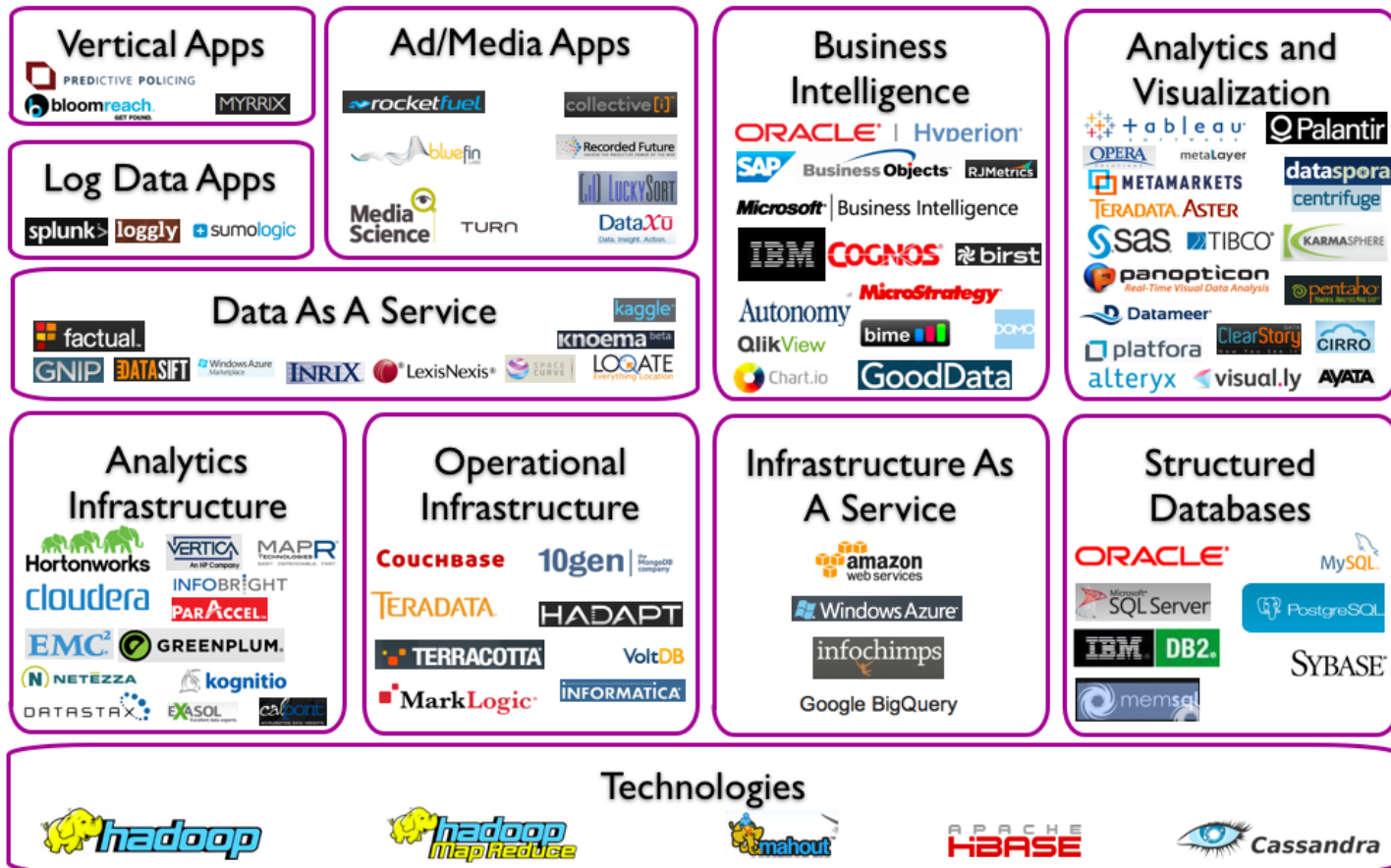
Poor data quality costs the US economy around

\$3.1 TRILLION A YEAR

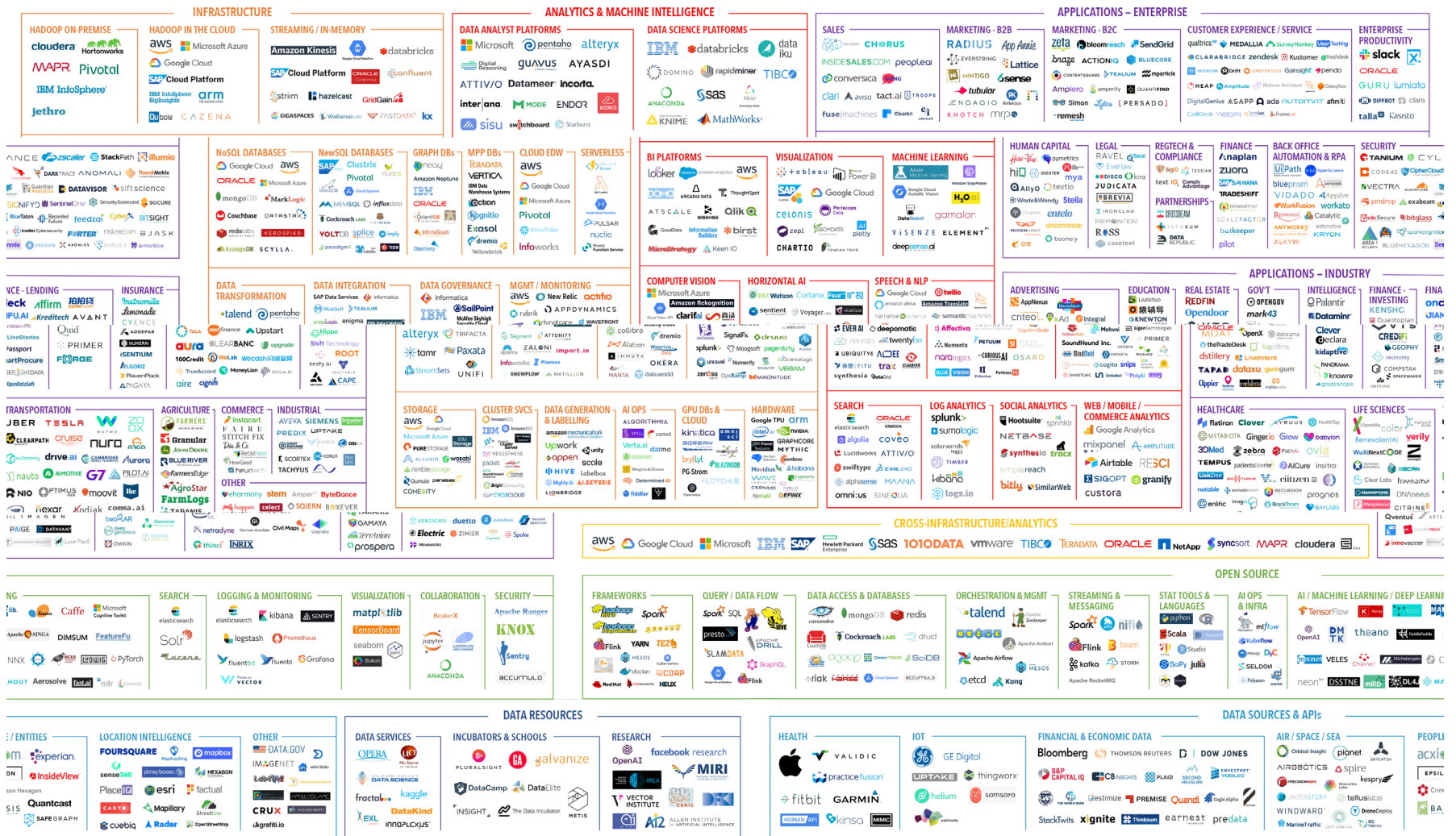


Big data landscape (2012)

Big Data Landscape



Big data landscape (Now)





Data contains value and knowledge

Data Processing & Mining

- But to extract the knowledge, data needs to be
 - Stored (managed)
 - Processed
 - Analyzed
- Diagram illustrating the flow of data processing and mining:
- Data Engineering → Big Data Processing
 - Data Analysis → Data Mining

Big Data vs. Data Science vs. Data Mining vs. Data Engineering ??

What is Data Mining?

- Given lots of data
- Discover patterns and models that are:
 - **Valid:** hold on new data with some certainty
 - **Useful:** should be possible to act on the item
 - **Unexpected:** non-obvious to the system
 - **Understandable:** humans should be able to interpret the pattern
- Data mining: knowledge discovery from data

Data Mining Tasks

- **Descriptive methods**

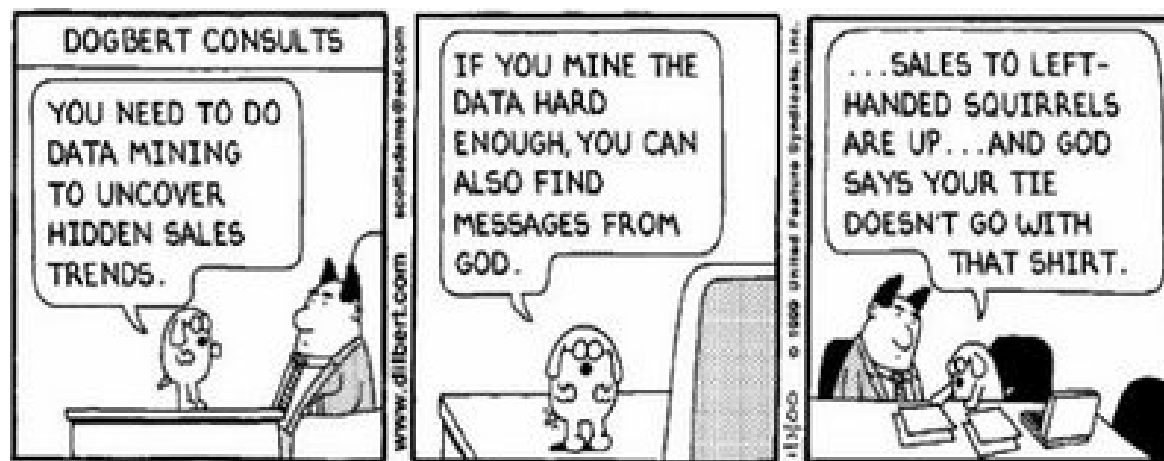
- Find human-interpretable patterns that describe the data
 - **Example:** Clustering

- **Predictive methods**

- Use some variables to predict unknown or future values of other variables
 - **Example:** Recommender systems

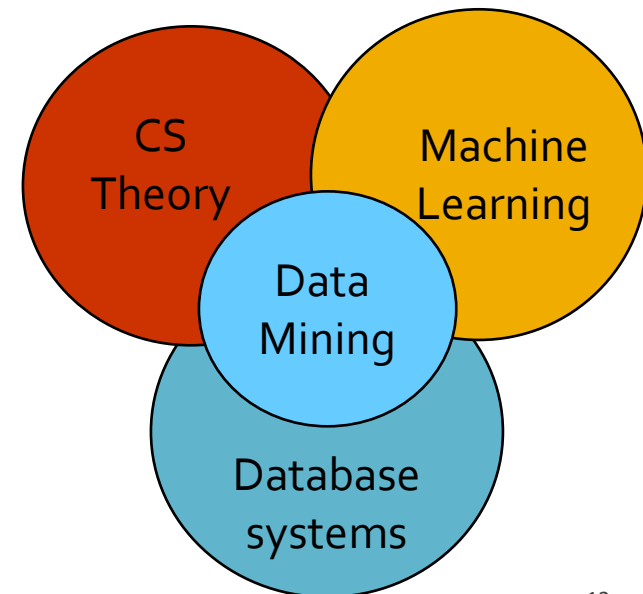
Meaningfulness of Analytic Answers

- A risk with “Data mining” is that an analyst can “discover” patterns that are meaningless
- Statisticians call it **Bonferroni’s principle**:
 - Roughly, if you look in more places for interesting patterns than your amount of data will support, you are bound to find crap



Data Mining: Cultures

- **Data mining overlaps with:**
 - **Databases:** Large-scale data, simple queries
 - **Machine learning:** Large data, complex models
 - **CS Theory:** (Randomized) Algorithms
- **Different cultures:**
 - To a DB person, data mining is an extreme form of **analytic processing** – queries that examine large amounts of data
 - Result is the query answer
 - To a ML person, data-mining is the **inference of models**
 - Result is the parameters of the model



This Class

- This class overlaps with machine learning, statistics, artificial intelligence, databases but more stress on
 - Algorithms/theory
 - Data mining
 - Clustering, stream data
 - Computing architectures/model
 - Distributed computing (MapReduce, cloud computing)
 - In-storage computing (ISC)
 - Big data software platforms & eco-systems
 - Apache Hadoop/Spark
 - NoSQL/Key-Value Store (KVS)

Questions?
